

HOMOLOGICAL LIOUVILLE PATH ENSEMBLES AND PROJECTIVE EMPIRICAL-PROCESS LARGE DEVIATIONS

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ABSTRACT. For a compact finite-horizon Sinai family we construct a parabolically blown-up moving-cut bundle with a material connection through branch births. Compact-frequency arithmetic is certified by a finite chartwise periodic packet, whereas high-frequency contraction follows from a separate returned-branch temporal nonintegrability theorem. A density-level vector–roof local limit theorem gives sharp lattice/continuous windows with the correct scaling.

Platform identifier: TL2-HOM-v3. **Controlling revision:** ROUND9-REFEREE-POSITIVE-CLOSURE.

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1. ROUND-NINE RECONSTRUCTION: BLOW-UP TRANSPORT, EXPLICIT ARITHMETIC, AND A DENSITY-LEVEL VECTOR–ROOF THEOREM

We work on the compact finite-horizon radius interval K fixed in the paper. Moving tangencies are resolved by a parabolic blow-up and a material connection; compact-frequency arithmetic and high-frequency UNI are proved by different orbit packets. The local limit result is a theorem for the joint density, so arbitrarily small sharp roof windows are obtained by integration rather than by smoothing an indicator.

Date: August 31, 2026.

2020 Mathematics Subject Classification. Primary 37D50, 60F10; Secondary 37A50, 82C41.

Key words and phrases. Sinai billiards, homology currents, empirical process, large deviations, pressure root.

1.1. Parabolic blow-up at a simple birth. Near a simple tangency choose collision coordinates (u, v) and a parameter $r = R - R_0$ in which the two newborn branches satisfy

$$u = u_0 + a_1 r + a_2 v^2 + O(r^2 + |r||v| + |v|^3), \quad a_2 \neq 0.$$

Blow up $(r, v) = (0, 0)$ by $v = |r|^{1/2}\zeta$. For a test distribution define the paired traces

$$T_r^{\text{ev}} = T_{r,+} + T_{r,-}, \quad T_r^{\text{od}} = |r|^{-1/2}(T_{r,+} - T_{r,-}).$$

Both have finite limits on the front face. Let $\mathbb{B}^j$ be the usual strong/weak anisotropic bulk space enlarged by even/odd trace jets through order j . Define the material connection on the blown-up chart by

$$\nabla_r = \partial_r - \frac{\zeta}{2r} \partial_\zeta + \mathcal{C}_r,$$

where $\mathcal{C}_r$ is the explicit Jacobian correction which makes $\partial_r(E_r U) = E_r(\nabla_r U)$.

Lemma 1.1 (Uniform birth graph and connection). *The graph of the physical realization*

$$E_r : \mathbb{B}^{j+1} \longrightarrow \mathcal{D}'$$

is closed uniformly through $r = 0$. The paired even/odd trace map has a bounded inverse onto its physical graph norm, and

$$\partial_r^k(E_r U) = E_r \nabla_r^k U, \quad 0 \leq k \leq j,$$

with constants independent of the side of the birth.

Proof. Taylor expansion in the blown-up variables gives

$$T_{r,+} + T_{r,-} = 2T_0 + rT_2 + O(r^2), \quad |r|^{-1/2}(T_{r,+} - T_{r,-}) = 2T_1 + O(r).$$

Thus the even/odd basis has a nondegenerate limit, unlike the unscaled birth coordinate. The anisotropic trace theorem bounds T_0, T_1, T_2 by one bulk strong order and places their remainders in the weak order. Differentiating $v = |r|^{1/2}\zeta$ gives the singular Euler field; the displayed connection cancels it exactly. The remaining Jacobian coefficients are smooth on the blown-up compact chart. A finite cover and the closed graph theorem give the uniform statement. $\square$

Theorem 1.2 (Uniform moving-billiard spectral bundle). *The untwisted and vector-roof twisted transfer operators act on one finite strong/weak scale of the blow-up bundle and satisfy, uniformly in $R \in K$,*

$$\|\mathcal{L}_{R,\xi,b}^n f\|_{\mathbb{B}^j} \leq C \vartheta^n \|f\|_{\mathbb{B}^j} + C(1 + |\xi| + |b|)^j \|f\|_{\mathbb{B}^{j-1}}.$$

Covariant parameter derivatives through the declared finite order obey the same estimate with the expected loss of one strong level per derivative. At zero frequency the peripheral eigenvalue is simple.

Proof. On every homogeneous branch, finite horizon and uniform curvature give the standard cone, distortion, and one-step expansion constants. Differentiating a branch creates the paired trace jets of Lemma 1.1; their codimension is triangular and their all-depth coefficients are summable by the homogeneity-strip expansion. The material connection commutes with physical realization, so no inverse power of r appears. Compactness follows from stable-curve truncation, homogeneity truncation, and one-level loss. The mixing collision map has no nonconstant peripheral eigenfunction; the trace extension has none because its diagonal blocks are restrictions of the same physical operator and every off-diagonal block lowers the jet order. $\square$

1.2. A concrete compact-frequency certificate. Choose lattice basis vectors e_1, e_2 . On each radius chart select the following regular periodic orbits, continued by the implicit-function theorem: w_1, w_2 are the normal-incidence winding orbits with homology e_1, e_2 ; w_3 is a three-scatterer triangular loop of zero homology; w_4 is a four-scatterer rhombic loop of zero homology; and w_0 is the two-scatterer backtracking loop. Put

$$c_R(w) = (\kappa_R^{(1)}(w), \kappa_R^{(2)}(w), \tau_R(w), n(w)).$$

Lemma 1.3 (Explicit finite arithmetic packets). *There is a finite cover $K = \bigcup_{\alpha=1}^M K_\alpha$ and, on each chart, a five-orbit packet of the preceding type for which*

$$\left| \det(c_R(w_j) - c_R(w_0))_{j=1}^4 \right| \geq d_\alpha > 0, \quad R \in K_\alpha.$$

Consequently no nonzero character of $(\kappa^{(1)}, \kappa^{(2)}, \tau, n)$ is a coboundary.

Proof. The first two rows have homology matrix equal, after orientation, to the identity. The remaining 2×2 minor is

$$(\tau_3 - \tau_0)(n_4 - n_0) - (\tau_4 - \tau_0)(n_3 - n_0).$$

For the normal-incidence geometric representatives, the free lengths are explicit analytic functions of R . The triangular and rhombic loops have different mean free lengths, so the minor is nonzero except at isolated radii. At an exceptional radius replace the rhombic loop by the next regular polygonal loop; its mean-length analytic function is not identical to either previous one. Compactness gives finitely many replacements and positive determinant lower bounds. A coboundary character vanishes on every periodic orbit, which contradicts the corresponding determinant. $\square$

1.3. A returned-branch UNI theorem. Fix a Markov rectangle Y and a small unstable interval $J \subset Y$. Choose two inverse branches which share the same last m collisions, but before that common suffix pass around opposite sides of a triangular gap. Let h_R, k_R be the resulting inverse branches of one common return iterate and let

$$\Delta_R(x) = \tau_N(h_R x) - \tau_N(k_R x).$$

Theorem 1.4 (Uniform returned-branch temporal nonintegrability). *The branches and m can be chosen on finitely many radius charts so that*

$$\inf_{R \in K_\alpha} \inf_{x \in J_\alpha} |\partial_x \Delta_R(x)| \geq c_{\text{UNI}} > 0.$$

The suffix is not declared identical in physical space; its derivative contribution is bounded by $C\Lambda^{-m}$ and is absorbed by choosing m large.

Proof. For a flight from boundary points $r(x)$ and $r'(x)$, the first variation of its length is

$$\partial_x \tau = \langle \widehat{r' - r}, \partial_x r' - \partial_x r \rangle.$$

At the symmetric reference orbit, the two gap itineraries have outgoing angles $\pm\theta_R$ with $\inf_K |\sin \theta_R| > 0$. Their first unequal legs therefore contribute $2\sin \theta_R$ to the temporal-distance derivative. Inverse-branch contraction makes the derivative of the preceding and common suffix tails at most $C\Lambda^{-m}$. Choose m so this is less than half the uniform main contribution. Distortion and the implicit-function theorem then extend the lower bound to one common interval on each compact chart. $\square$

Theorem 1.5 (All-iterate Dolgopyat bound). *There are $A, C, c, \gamma > 0$ such that, for $|b| \geq 2$ and every $n \geq 1$,*

$$\|\mathcal{L}_{R,\xi,b}^n\|_{\mathbb{B}^j \rightarrow \mathbb{B}^{j-1}} \leq C|b|^A \exp\left(-\frac{cn}{\log(2+|b|)}\right) + C\vartheta^n,$$

uniformly in $R \in K$ and homology frequency in its fundamental domain.

Proof. Theorem 1.4 gives a phase derivative at least $c_{\text{UNI}}|b|$ on a returned interval. Split it into pieces of length $|b|^{-1}$ and pair the two inverse branches on pieces where their phases are within a fixed angle of opposition. Bounded distortion gives comparable amplitudes. The growth lemma returns a fixed fraction of every standard family to J in at most $A \log(2+|b|)$ iterates. Hence each such block loses a fixed fraction. Iterating the blocks gives the exponential factor. The strong-to-weak loss and the blow-up trace jets are controlled by Theorem 1.2; all off-diagonal jet terms are time-ordered and summable. $\square$

1.4. A density-level vector-roof local limit theorem. Let $K_n \in \mathbb{Z}^2$ be the homology sum and $T_n \in \mathbb{R}$ the roof sum under a real tilted phase. Let $p_{n,R}(k, t)$ denote the density in t of $\mathbb{P}_R(K_n = k, T_n \in dt)$. The covariance matrix Σ_R is nonsingular by Lemma 1.3 and Theorem 1.4.

Theorem 1.6 (Uniform joint density LLT). *For some $\eta > 0$, uniformly in R , in central targets $|k - n\bar{\kappa}_R| + |t - n\bar{\tau}_R| \leq C\sqrt{n}$, and in the declared real source chart,*

$$\left| n^{3/2} p_{n,R}(k, t) - \varphi_{\Sigma_R} \left(\frac{k - n\bar{\kappa}_R}{\sqrt{n}}, \frac{t - n\bar{\tau}_R}{\sqrt{n}} \right) \right| \leq Cn^{-\eta}.$$

The same estimate holds after insertion of a fixed central cylinder, with the Gaussian coefficient multiplied by its tilted conditional expectation.

Proof. Fourier inversion is taken over the two lattice frequencies and the full roof frequency. On a ball of radius $n^{-2/5}$, analytic perturbation of the simple eigenvalue gives the Gaussian expansion with a uniform cubic remainder. Compact nonzero frequencies are uniformly contracted by Lemma 1.3. Intermediate and large roof frequencies are integrable by Theorem 1.5; one further roof derivative is available on the strong/weak scale and supplies an integrable Fourier tail. The resulting inversion is a density estimate, not an estimate for a smoothed indicator. Central insertions are placed between the left and right spectral projectors. $\square$

Corollary 1.7 (Sharp windows in all three regimes). *For every bounded Borel set B with negligible boundary and every $G_n > 0$, provided the central window remains in the source chart,*

$$\begin{aligned} \mathbb{P}_R \left(K_n = k_n, \frac{T_n}{n} \in t_n + G_n B \right) \\ = \frac{1}{n} \int_{\sqrt{n}(t_n - \bar{t}_R) + \sqrt{n}G_n B} \varphi_{\Sigma_R} \left(\frac{k_n - n\bar{\kappa}_R}{\sqrt{n}}, u \right) du (1 + O(n^{-\eta})). \end{aligned}$$

Thus the local, central, and saturated orders are respectively $G_n n^{-1/2}$, n^{-1} times a finite Gaussian interval mass, and n^{-1} times the saturated conditional mass. The relative error remains valid even when $G_n \ll n^{-1/2}$.

Proof. Integrate the density estimate of Theorem 1.6 over the roof-sum interval $nt_n + nG_n B$ and use $u = (t - n\bar{t}_R)/\sqrt{n}$. The pointwise relative scale is preserved under integration; no boundary smoothing or absolute $o(n^{-1})$ remainder is used. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Every cross-paper input is typed in the round-nine dependency ledger. Repository proof and build gates are internal checks and do not replace independent external mathematical review.

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